

DaSEH Day 5 Cheatsheet

Data Cleaning

Major concepts

- **Most important rule of data handling:** Always look at your data.
- **NA:** General missing data.
- **NaN:** “Not a Number”; occurs when you calculate 0/0.
- **Inf and -Inf:** Infinity; occurs when you divide a positive or negative number by 0.
- **naniar:** Useful for exploring missingness in large datasets. See the [naniar package](#).
- **tidyverse (stringr):** Useful for detecting and working with parts of character values, especially with `filter()` and `str_detect()`.

Examples

Example of `case_when()` with `mutate()`:

```
Orange |>
  mutate(old = case_when(
    age > 1000 ~ TRUE,
    .default = FALSE
  ))
```

Functions

Library/Package	Piece of code	Example of usage	What it does
Base R	<code>is.na()</code>	<code>is.na(x)</code> <code>any(is.na(x))</code>	Checks if <code>x</code> is NA. With <code>any()</code> , checks if any values are NA.
Base R	<code>is.nan()</code>	<code>is.nan(x)</code>	Checks if <code>x</code> is NaN.
Base R	<code>is.infinite()</code>	<code>is.infinite(x)</code>	Checks if <code>x</code> is Inf or -Inf.
naniar	<code>pct_complete()</code>	<code>pct_complete(x)</code>	Reports the percentage of complete data in <code>x</code> .

Library/Package	Piece of code	Example of usage	What it does
naniar	<code>gg_miss_var()</code>	<code>gg_miss_var(x)</code>	Plots the percentage of missing data by variable in <code>x</code> .
naniar	<code>miss_var_summary()</code>	<code>miss_var_summary(x)</code>	Reports the percentage of missing data by variable in <code>x</code> .
naniar	<code>miss_var_which()</code>	<code>miss_var_which(x)</code>	Returns variable names that contain missing values in <code>x</code> .
tidyverse (tidyr)	<code>drop_na(df)</code>	<code>drop_na(df)</code> <code>drop_na(variable)</code>	Drops rows with NA values. When a variable is specified, only drops rows where that variable is NA.
tidyverse (dplyr)	<code>na_if()</code>	<code>na_if(vector, value)</code>	Converts specified values to NA. Use with <code>mutate()</code> .
tidyverse (tidyr)	<code>replace_na()</code>	<code>replace_na(vector, value)</code>	Replaces NA values with a specified value. Use with <code>mutate()</code> .
tidyverse (dplyr)	<code>case_when()</code>	<code>case_when(test ~ outcome, .default = x)</code>	Vectorizes multiple <code>if_else()</code> statements. If no cases match, NA is returned unless <code>.default</code> is specified. Use with <code>mutate()</code> .
tidyverse (dplyr)	<code>mutate()</code>	<code>df <- mutate(df, newcol = wt / 2.2)</code>	Adds a new column based on existing columns.
tidyverse (tidyr)	<code>separate()</code>	<code>df > separate(x, c("A", "B"))</code>	Separates one character column into multiple columns using a regular expression or numeric locations.
tidyverse (tidyr)	<code>unite()</code>	<code>df > unite("z", x:y, remove = FALSE)</code>	Unites multiple columns into one column.
tidyverse (stringr)	<code>str_detect()</code>	<code>df > filter(str_detect(col_name, "string_pattern"))</code>	Returns a logical vector indicating whether a string pattern was detected.
tidyverse (stringr)	<code>str_replace()</code>	<code>str_replace(string = vector, "replace_me", "with_me")</code>	Replaces the first instance of one specified string with another.
tidyverse (stringr)	<code>str_replace_all()</code>	<code>str_replace_all(string = vector, "replace_me", "with_me")</code>	Replaces all instances of one specified string with another.
tidyverse (stringr)	<code>str_sub()</code>	<code>str_sub(string = vector, start = 1, end = 3)</code>	Subsets a string to the specified character positions.

* This cheatsheet format was adapted from Alex's Lemonade Stand materials ([source](#)).

* Find more resources at <https://daseh.org>.